

QUESTION :

A rigid tank is divided into two compartments by a partition. One compartment contains 3 kmol of N_2 at 600 kPa and the other compartment contains 7 kmol of CO_2 at 200 kPa. The partition is removed, and the two gases form a homogeneous mixture at 300 kPa.

The partial pressure of N_2 in the mixture is

Option	Partial Pressure of N_2
(A)	75 kPa
(B)	90 kPa
(C)	150 kPa
(D)	3175 kPa

ANSWER :

Mole Fraction of N_2

The total number of moles in the mixture is

$$n_{\text{total}} = n_{N_2} + n_{CO_2}$$

$$n_{\text{total}} = 3 + 7 = 10 \text{ kmol}$$

Therefore, the mole fraction of N_2 is

$$y_{N_2} = \frac{n_{N_2}}{n_{\text{total}}}$$

$$y_{N_2} = \frac{3}{10} = 0.3$$

Partial Pressure

For an ideal gas mixture, the partial pressure is given by

$$P_{N_2} = y_{N_2} P_{\text{mixture}}$$

$$P_{N_2} = 0.3 \times 300$$

$$P_{N_2} = 90 \text{ kPa}$$

Hence, the correct option is

$$(B) \text{ 90 kPa}$$

Derivation of Partial Pressure Relation

For the entire ideal gas mixture,

$$P_{\text{mixture}}V = n_{\text{total}}RT$$

For component i, its partial pressure is defined as the pressure exerted by component i if it alone occupied the entire volume V at the same temperature T.

Therefore,

$$P_iV = n_iRT$$

Dividing the above two equations,

$$\frac{P_iV}{P_{\text{mixture}}V} = \frac{n_iRT}{n_{\text{total}}RT}$$

Canceling V, R, and T,

$$\frac{P_i}{P_{\text{mixture}}} = \frac{n_i}{n_{\text{total}}}$$

The mole fraction of component i is defined as

$$y_i = \frac{n_i}{n_{\text{total}}}$$

Hence,

$$P_i = y_i P_{\text{mixture}}$$

Therefore, for nitrogen,

$$P_{N_2} = y_{N_2} P_{\text{mixture}}$$

QUESTION :

An 80 L rigid tank contains an ideal-gas mixture of 5 g of N_2 and 5 g of CO_2 at a specified pressure and temperature. If N_2 were separated from the mixture and stored at the mixture temperature and pressure, its volume would be

Option	Volume
(A)	32 L
(B)	36 L
(C)	40 L
(D)	49 L

ANSWER :

Number of moles of each gas

For N_2 ,

$$n_{N_2} = \frac{m_{N_2}}{M_{N_2}} = \frac{5}{28}$$

For CO_2 ,

$$n_{CO_2} = \frac{m_{CO_2}}{M_{CO_2}} = \frac{5}{44}$$

The total number of moles is

$$n_{total} = \frac{5}{28} + \frac{5}{44}$$

Mole Fraction of N_2

The mole fraction of N_2 is

$$y_{N_2} = \frac{n_{N_2}}{n_{total}}$$
$$y_{N_2} = \frac{\frac{5}{28}}{\frac{5}{28} + \frac{5}{44}} = \frac{44}{72} = \frac{11}{18}$$

Volume of N_2

At the same temperature and pressure, the volume occupied by a component is proportional to its number of moles. Therefore,

$$\frac{V_{N_2}}{V_{mixture}} = \frac{n_{N_2}}{n_{total}} = y_{N_2}$$

Hence,

$$V_{N_2} = y_{N_2} V_{mixture}$$

$$V_{N_2} = \frac{11}{18} \times 80$$

$$V_{N_2} = 48.89 \text{ L} \approx 49 \text{ L}$$

Hence, the correct option is

(D) 49 L

QUESTION :

An ideal-gas mixture consists of 3 kg of Ar and 6 kg of CO₂ gases. The mixture is now heated at constant volume from 250 K to 350 K. The amount of heat transfer is

Option	Heat Transfer
(A)	374 kJ
(B)	436 kJ
(C)	488 kJ
(D)	525 kJ

ANSWER :

Heat Transfer at Constant Volume

For a closed system, the first law of thermodynamics is

$$Q - W = \Delta U$$

Since the volume is constant, the boundary work is zero:

$$W = \int P dV = 0$$

Therefore,

$$Q = \Delta U$$

For an ideal-gas mixture,

$$\Delta U = m_{Ar} C_{v,Ar} \Delta T + m_{CO_2} C_{v,CO_2} \Delta T$$

The temperature change is

$$\Delta T = 350 - 250 = 100 \text{ K}$$

Using

$$C_{v,Ar} = 0.312 \text{ kJ/(kg} \cdot \text{K)}$$

and

$$C_{v,CO_2} = 0.657 \text{ kJ/(kg} \cdot \text{K)}$$

we get

$$Q = (3)(0.312)(100) + (6)(0.657)(100)$$

$$Q = 93.6 + 394.2$$

$$Q = 487.8 \text{ kJ} \approx 488 \text{ kJ}$$

Hence, the correct option is

(C) 488 kJ

QUESTION :

An ideal-gas mixture consists of 30 percent Helium and 70 percent Argon gases by mass. The mixture is now expanded reversibly and adiabatically in a turbine from 400°C and 1.2 MPa to a pressure of 200 kPa. The mixture temperature at turbine exit is

Option	Temperature
(A)	195°C
(B)	56°C
(C)	112°C
(D)	130°C

ANSWER :

Specific Heat Ratio of Mixture

Both Helium and Argon are monatomic gases. Therefore,

$$\gamma_{\text{He}} = \gamma_{\text{Ar}} = \frac{5}{3}$$

Hence, the specific heat ratio of the mixture is also

$$\gamma_{\text{mix}} = \frac{5}{3}$$

Temperature-Pressure Relation

For a reversible adiabatic process involving an ideal gas,

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}}$$

The initial temperature is

$$T_1 = 400 + 273 = 673 \text{ K}$$

The pressure ratio is

$$\frac{P_2}{P_1} = \frac{200}{1200} = \frac{1}{6}$$

For $\gamma = \frac{5}{3}$,

$$\frac{\gamma - 1}{\gamma} = \frac{\frac{5}{3} - 1}{\frac{5}{3}} = \frac{2}{5} = 0.4$$

Therefore,

$$T_2 = 673 \left(\frac{1}{6} \right)^{0.4}$$

$$T_2 \approx 329 \text{ K}$$

Converting to Celsius,

$$T_2 = 329 - 273$$

$$T_2 \approx 56^\circ\text{C}$$

Hence, the correct option is

$$(B) 56^\circ\text{C}$$